

Technical *L*ands

editors

Jeffrey S Nesbit
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A Critical *P*rim*er*

Technical lands are spaces united by their “exceptional” status – their remote location, delimited boundary, secured accessibility, and vigilant management. Designating land as “technical” is thus a political act. Doing so entails dividing, marginalizing, and rendering portions of the Earth inaccessible and invisible. An anti-visibility of technical lands enables forms of hypervisibility and surveillance through the rhetorical veil of technology. Including the political and physical boundaries, technical lands are used in highly aestheticized geographies to resist debate surrounding production and governance. These critical sites and spaces range from disaster exclusion and demilitarized zones to prison yards, industrial extraction sites, airports, and spaceports. The identification and instrumentalization of technical lands have increased in scale and complexity since the rise of neoliberalization. Yet, the precise theoretical contours that define these geographies remain unclear. *Technical Lands: A Critical Primer* brings together authors from a diverse array of disciplines, geographies, and epistemologies to interrogate and theorize the meaning and increasing significance of technical lands.

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Uncommon *Interests*

Rania Ghosn

In 2008, the oil and gas company Total (now Total Energies) conducted a corporate advertising campaign to articulate its position on the future of energy. This campaign was timed vis-à-vis the UN Intergovernmental Panel on Climate Change recommendations to reduce carbon emissions.¹ Under the motto *Communauté d'Intérêts* (Common Interests), the corporation's media campaign proposed to reconcile the growth objectives of private capital with climate challenges in a slogan that rhetorically articulated, "What if meeting energy demand and combating global warming are inseparable?"² In this campaign, Common Interests deployed a visual binary structure with sites of extraction on one side and consumption on the other. For example, one frame featured a prospective image of a polar glacier reflected into the iconic nighttime glimmer of Manhattan's skyline. What is eclipsed from these images is the technological system, including the myriad of tankers, pipes, and terminal ports that connect both geographies. Economists describe these discrepancies with the term "externalities." These externalities are the side effects of the private processes associated with allocating resources. They produce ill effects, notably pollution and the degradation of the commons. To paraphrase the economist William Kapp, who abstracted a general law for capitalism in his 1950 book *The Social Costs of Private Enterprise*, "Capitalism [read here as technological lands] must be regarded as an economy of unpaid costs, 'unpaid' in so far as a substantial portion of the actual costs of production remains unaccounted for in entrepreneurial outlays; instead, they are shifted to, and ultimately borne by, third persons or by the community as a whole."³ If technological lands do not matter, then energy corporations are not accountable or obligated to pay for the disasters brought about by extractivism.

Technological lands are not inadvertently omitted; they are excised by design. Their "designed" abstraction severs geography into a flattened binary of crude and refined lands, concealing frictions and externalities throughout the system. Indeed, the mandate for "clean" Manhattanism rests on the abstraction of technological lands. Hence, the crisis of Manhattanism (aka the climate crisis) is a crisis of the representation of technical lands. Accordingly, the climate crisis cannot be resolved by introducing more nature into either side of the frame. As Bruno Latour elaborates in the conclusion of *The Politics of Nature*, this is because "nature is not a particular sphere of reality but the result of such a political division."⁴ An ecological, counter-Manhattanist approach to the climate crisis addresses the politics of representation that underpin the division line itself and in so doing breaches the closure of "common interests."

The abstraction of technological lands is sustained by a nearly unbroken attitude of wonderment, or "energy myths," extending from the advent of steam power through the spread of fossil fuels. Not dissimilar to John Gast's 1872 "American Progress," technological infrastructures, such as telegraph lines, railroads, and steamships, came to embody visions of progress. Such technical "fixes" reproduce a series of what historian George Basalla calls "energy myths," in which any newly discovered source of energy is "assumed

¹ <https://totalenergies.com/media/news/news/campagne-institutionnelle-2008-communaute-dinterets>

² "Corporate advertising campaign: 'Common interests,'" Total, May 5, 2008, <https://www.total.com/en/media/news/news/campagne-institutionnelle-2008-communaute-dinterets>.

³ William Kapp, *The Social Costs of Private Enterprise* (New York, NY: Schocken Books, 1971), 231.

⁴ Bruno Latour, *Politics of Nature: How to Bring the Sciences into Democracy* (Cambridge, MA: Harvard University Press, 2004), 231–32.

⁵ George Basalla, "Some Persistent Energy Myths," *Energy and Transport: Historical Perspectives on Policy Issues*, eds. George H. Daniels and Mark H. Rose (Beverly Hills, CA: Sage, 1982), 27–28.

to be without faults, infinitely abundant, and to have the potential to affect utopian changes in society. These myths persist until a new energy source is deployed to the point that its drawbacks become apparent."⁵ The proposed new source of energy, however, is not treated any differently. Rather, recently discarded energy myths are resuscitated and conferred upon the newcomer, replete with familiar images and narratives. These cyclical narratives do not acknowledge, let alone reform, technological lands. Instead, these myths reduce the question of energy transition to a problem of technology, the response to which is to overlook the obsolete and wretched grounds of fossil fuels and displace desire to another energy technology. It is essential to add that myths are not the gap between the technological and spatial, a disjunction that gives rise to misrepresentations that might be corrected with empirical evidence. Myths are geographic projects by design; they remove technical lands from representation and occupy that void with iconographic imagery.

The externalization of costs has been the explicit mandate of the oil industry. No energy myth is greater than the promise of infinite fossil fuels. As the largest single commodity in international trade, both in value and weight, oil has inscribed its infrastructural order as a mode of planetary organization.⁶ Over the past 200 years, particularly since World War II, the high-energy content of black gold has tied economic growth to promises of progress and development. Developmentalism praised oil as a mode of energy that could expand economies and advance democracies in producing and consuming nations alike. In social indicators, such as the United Nations Development Indicators, a nation's prosperity was equated with the rate of energy use per capita, contributing to the prodigious expansion of the fossil fuel industry.

The promises of fossil fuels came down with a disastrous twist. The 1970s marked the end of cheap, abundant, and guilt-free petroleum. A surge in global energy demands and the need to manage carbon dioxide have made the imperative for change in energy systems. The question of the transition inherently became a problem of carbon, the response to which was a series of techno-fixes—carbon sequestration, carbon credits, carbon markets—along with low-carbon energy sources from the sun, wind, and tides.

Representation is complicit, possibly foundational in perpetuating such technological myths. The energy myths of zero-carbon alternatives are uncannily familiar, switching off political vigilance on reorganizing the world and its values. AMO/OMA, for example, has explored new energy policies in a series of projects, such as *Roadmap 2050: A Practical Guide to a Prosperous, Low-Carbon Europe*, which proposed a diversified and regional energy network to reduce Europe's carbon-dioxide emissions. A series of renderings visualize how the integrated European network reconfigures the continent's geography away from political boundaries and into energy regions: Tidal States, Isles of Wind, Biomassburg, and Solaria, the latter of which brought together the Mediterranean sunbelt countries. The rendering "Parisian Energy from Sahara Sun" illustrates the promises of Solaria with the Eiffel Tower on one side of the Mediterranean and a camel caravan

⁶ Matthew T. Huber, *Lifelihood: Oil, Freedom, and the Forces of Capital* (Minneapolis, MN: University of Minnesota Press, 2013); Hannah Appel, Arthur Mason, and Michael Watts, eds. *Subterranean Estates: Life Worlds of Oil and Gas* (Ithaca, NY: Cornell University Press, 2015); Imre Szeman, *On Petrocultures: Globalization, Culture, and Energy* (Morgantown, WV: West Virginia University Press, 2019).

in a North African field of solar panels on the other. The Solaria symbolizes a promise of a radical transition in energy technology. Its visual structure—the metropolis on the one hand and the extractive hinterland on the other—is an all too familiar organizational trope, reminiscent of Total's Common Interests campaign. It perpetuates the binary organization of the carbon territory and further ignores the scorched, technical lands of oil and gas fields.

Representation is also necessary to dispel myths—to make technological lands visible and to make their externalities matter. In 2005, a time of ecological turmoil, Bruno Latour's *Making Things Public* examined the agency of representation in bringing things (including technologies) to the public's attention.⁷ Latour writes:

A matter of concern, is what happens to a matter of fact when you add to it its whole scenography, much like you would do by shifting your attention from the stage to the whole machinery of a theatre ... Instead of simply being there, matters of fact begin to look different, to render a different sound, they start to move in all directions, they overflow their boundaries, they include a complete set of new actors, they reveal the flimsy envelopes in which they are housed.⁸

Matters of concern also accept contingency; they are political *because* they are open to multiple, contradictory interests. Matters of concern demystify a picture-perfect image of progress and a singular politics of improvement and present ongoing contested uncertainties. In this world view, the role of representation shifts from reinforcing consensus on matters of fact to assembling public activism around controversies and other matters of concern.

An "architecture with externalities" counteracts these energy myths by delaminating technical lands and drawing out the scope of externalities *within* the world. Contrary to the nineteenth-century thermodynamic concept of energy, which erased qualitative differences in favor of exchange value, the renewed imagination sees technological externalities as *within* the lands.⁹ In this world view, geography is constantly re-formed by the oil network. Timothy Mitchell notes that closely following the oil entrails "tracing the connections that were made between pipelines and pumping stations, refineries and shipping routes, road systems and automobile cultures, dollar flows and economic knowledge, weapons experts and militarism." His analysis suggests a framing of the oil network across the boundaries between the material and the ideal, the political and the cultural, and the natural and the social.¹⁰

The representational challenges remain acute because of the narrative and strategic challenges posed by the relative invisibility of the climate crisis and its delayed effects. As Rob Nixon asks in his book *Slow Violence and the Environmentalism of the Poor*, how can we convert the slow-moving disasters that are long in the making into arresting stories, images, and symbols?¹¹ What is the descriptive machine that makes technological lands visible, political, and speculative? The next challenge for architects, landscape architects, and

⁷ Bruno Latour and Peter Weibel, eds., *Making Things Public: Atmospheres of Democracy* (Cambridge, MA: MIT Press, 2005).

⁸ Bruno Latour, *What Is the Style of Matters of Concern?* (Spinoza lectures, University of Amsterdam, Department of Philosophy, April–May 2005), 39, <http://bruno-latour.fr/node/16/>.

⁹ In *New Geographies 2: Landscapes of Energy*, I examined and articulated the emergence of the geographic—then a new, but for the most part latent, paradigm in design—and brought it to bear on the agency of design, energy, and space. The issue invited designers to take on the synthesizing role that geography aspired to play among the physical, economic, sociopolitical, and representational, with the invitation to engage "energy as a spatial project."

¹⁰ Timothy Mitchell, "Carbon Democracy," *Economy and Society* 38, no. 3 (2009): 422.

¹¹ Rob Nixon, *Slow Violence and the Environmentalism of the Poor* (Cambridge, MA: Harvard University Press, 2011).

urban designers might be to articulate and deploy a geographic representational machine—more fantastic than the eco-mimetic render—that visualizes externalities and illustrates the relations between form and politics that underpin the organization of all energy systems. A geographic imagination of energy unfolds the technological lands that were squashed into the division-line; it populates them with things deemed to be the mere fallout of progress; it repositions the new assembly at the center of political life.

To frame technological lands as a matter of concern, the research practice Design Earth deploys the architectural drawing and narrative as a method to visualize how technological systems change Earth and speculate ways of living with environmental technologies, such as oil fields and landfills, on a damaged planet.¹² The architectural project becomes the medium to foreground geopolitics that are situated, material, volumetric, populated, heterogeneous, and always grounded. The section drawing brings above- and belowground together in one frame, synthesizing the geological fixity of deep-time hydrocarbon fossilization with the surface order of modern material circulation and accumulation. Furthermore, the axonometric section situates oil reservoirs in a specific site while constructing a topological imagination of the planet. These descriptive geographic drawings unearth relational tensions between the situated and the planetary, the technological and the geographic.

Beyond documentary descriptions, the drawing engages a speculative narrative and sensibility, giving meaning to what otherwise would illustrate the unbearable scale of geopolitical events occurring around the globe, across time. It bestows meaning to facts. The narrative is a thought experiment, envisioning a possible future that opposes a stifling reality imposed by extractive planetary infrastructure. Rachel Carson, whose book *Silent Spring* is credited with launching the global environmental movement, refused to adhere to conventional methods of relaying scientific information. To give shape to the toxicity of pesticides, Carson resorted to using a narrative form. *Silent Spring* opens with “A Fable for Tomorrow,” which introduces a “town in the heart of America” that awakes to a birdless, budless spring. “This town does not actually exist,” Carson writes, “but it might easily have a thousand counterparts in America or elsewhere in the world,” for “a grim specter has crept upon us almost unnoticed, and this imagined tragedy may easily become a stark reality we all shall know.”¹³ Fabulation becomes a way to reframe reality beyond realism. It plots and gives figurative shape to threats whose repercussions are dispersed across space and time and draws public attention to such long-term, destructive acts.

A speculative drawing is a way to think and look at the carbon present, not as the phantom fulcrum between idealized pasts and salvific futures. The series of drawings recast technological lands with new mythologies that clarify the long-term consequences of the current fossil-fuel usage as the continued energy system on which modern life depends. It is vital to ask how designers speculate on technological lands when we can no longer imagine the world without technology and disaster. The drawings illustrate

¹² See Rania Ghosn and El Hadi Jazairy, *Geostories: Another Architecture for the Environment* (New York, NY: Actar, 2018). On the issue of oil, see Rania Ghosn and El Hadi Jazairy, “A Geographic Stroll around the Horizon,” *MONU* 20 (Spring 2014): 12–17; Rania Ghosn and El Hadi Jazairy, “Hassi Messaoud Oil Urbanism,” *New Geographies* 6, Grounding Metabolism, eds., Daniel Ibañez and Nikos Katsikis (November 2014): 144–53.

¹³ Rachel Carson, *Silent Spring* (Boston, MA: Houghton Mifflin, 1962), 3.

worlds in which “staying with the trouble”—what Donna Haraway says is both “to stir up potent response to devastating events, as well as to settle troubled waters”—is the only ethical option for technological and environmental matters.¹⁴

After Oil

How can speculative narratives reclaim the geographic imagination of energy—the forms, technologies, economies, and logistics of fossil fuels in architectural discourse and political life? Our project, *After Oil*, visualizes the formal embeddedness of carbon geographies to underline the politics of carbon reform. All energy transitions are no more formally neutral than the forms of carbon urbanism are politically neutral. In the *Geostories* series, *After Oil* emphasizes the agency of representation in the construction of carbon futures at a moment when the energy-economy-environment triad is at the forefront of design concerns. The project presents three speculative fictions on a future in which the Persian Gulf states and the world will transition away from fossil fuels as forms of energy. The *After Oil* drawings critically engage the present and future geographies of oil in the Gulf region, charting matters of concern for sites of extraction (Das Island), transit logistics (Strait of Hormuz), and the slow violence of climate change (Bubiyan Island). *After Oil* renders visible the embeddedness of the oil system in the Gulf region and invites us to imagine the long-range effects of such a crude relationship with Earth.

1. Das Island, Das Crude

Das Island is a critical Emirati offshore oil-and-gas industrial facility. Since the first oil-exploration expeditions in 1953, the island has fueled the urbanization of Dubai and Abu Dhabi, with many of the country’s iconic buildings being built with oil wealth. Das Crude makes the value displacement in oil urbanism visible by imagining the island with a subsurface field of depleted oil reservoirs. The United Arab Emirates’ architectural landmarks are indexed in relation to oil’s geological depths and times of extraction. The volume of excavated soil and stone is assembled in an artificial mountain, a landform monument to the age of oil.

fig 1–3

2. Strait of Hormuz Grand Chessboard

The Strait of Hormuz is a critical oil-transit chokepoint. Twenty percent of the world’s oil moves through the two-mile-wide shipping lane in its twenty-one-mile-wide passage. The strait has never actually been shut down, despite the persistent geopolitical anxiety over territorial disputes, particularly the disagreement between the UAE and Iran over the islands of Abu Musa, Greater Tunb, and Lesser Tunb. The Grand Chessboard repurposes the strait as a territorial real estate game, financed by the collective financial oil futures of traditional adversaries across the Gulf. The board game absorbs the three islands among the chess pieces of iconic speculative urban projects.

¹⁴ Donna Haraway, *Staying with the Trouble: Making Kin in the Chthulucene* (Durham, NC: Duke University Press, 2016), 1.



fig 1 Das Island, Das Crude. Design Earth, 2016

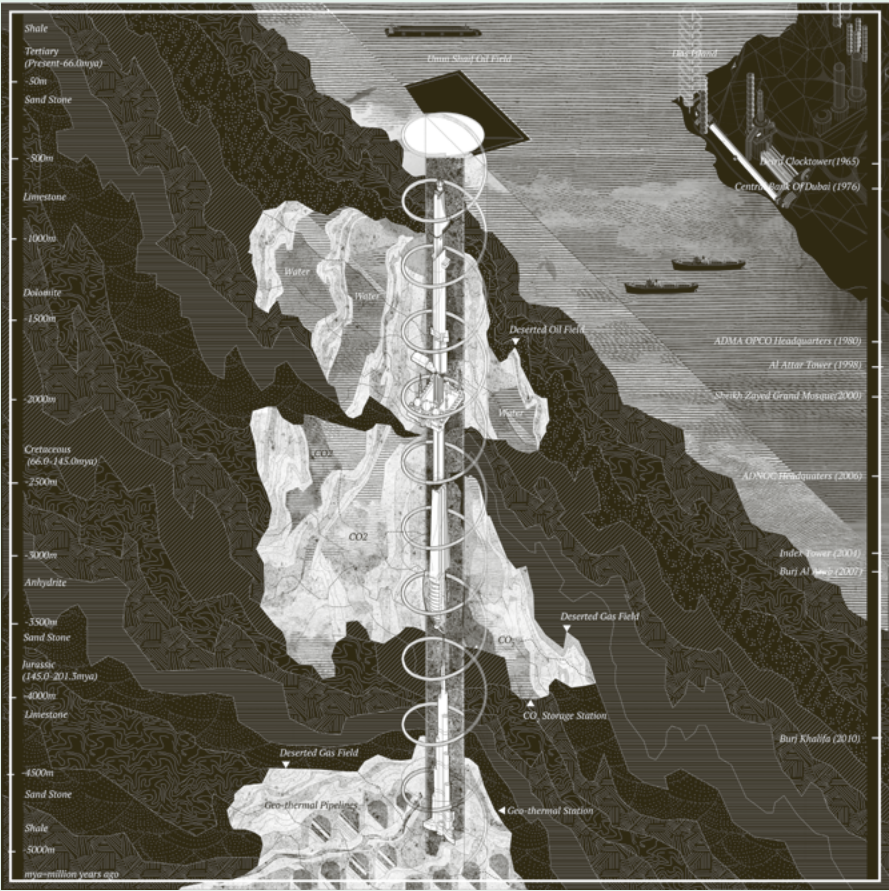


fig 2 Das Island, Das Crude. Design Earth, 2016

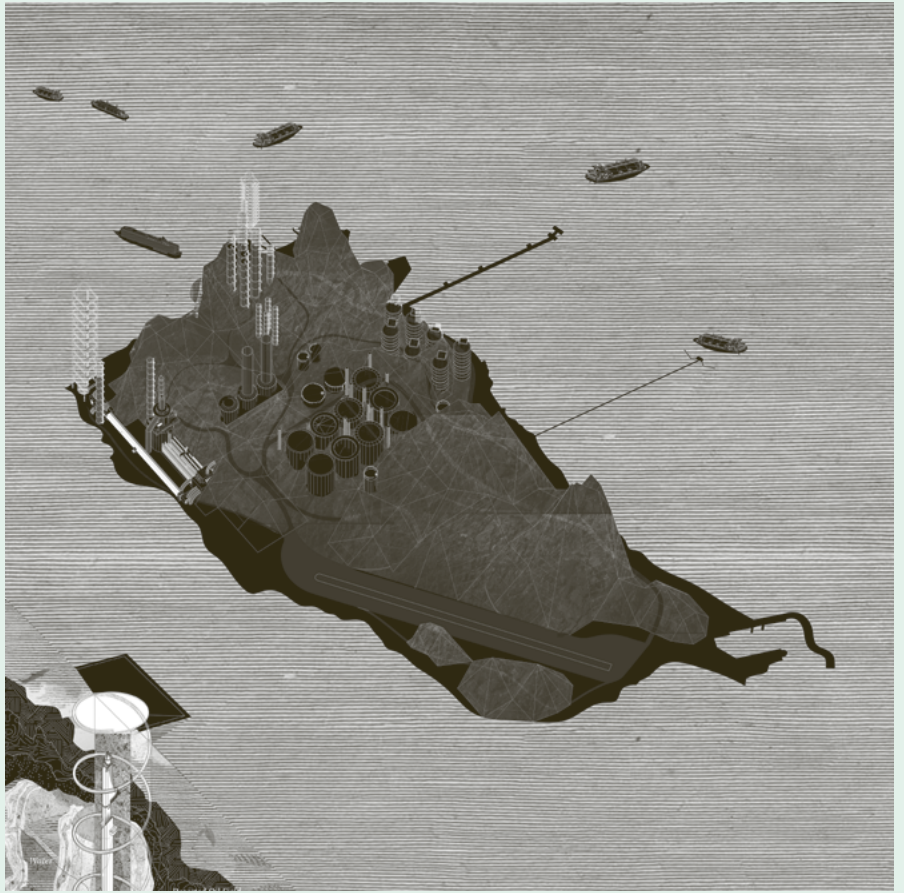


fig 3 Das Island, Das Crude. Design Earth, 2016

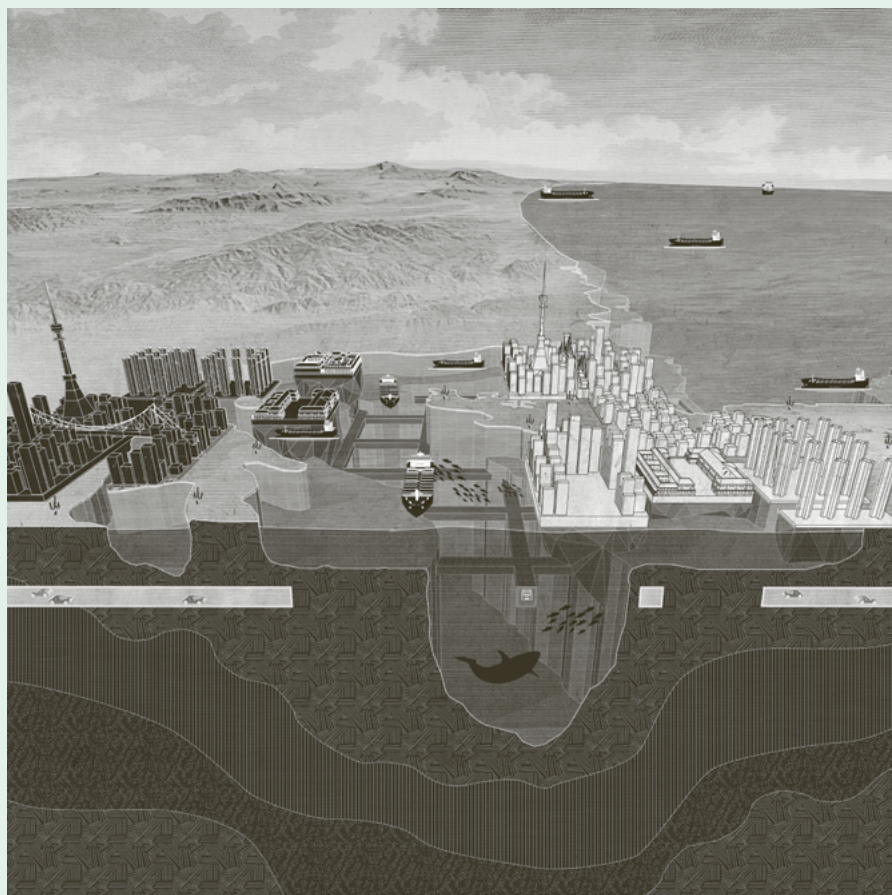


fig 4 Strait of Hormuz Grand Chessboard.
Design Earth, 2016



fig 5 Strait of Hormuz Grand Chessboard.
Design Earth, 2016

fig 4–6

3. Bubiyan Island, There Once Was an Island

The end of the Persian Gulf War in 1991 was accompanied by one of the largest oil spills, which drastically affected Kuwait's coastal environment. Despite the apocalyptic intensity of this geo-traumatic event, the business-as-usual oil industry, even with the increasing rate of carbon emissions, subjects the world to the slow violence of anthropogenic climate change. Bubiyan Island, flat and low-lying, is vulnerable to sea-level rise. The drawing, *There Once Was an Island*, shows this invisible threat by redrawing the island's shrinking shoreline and its sixteen highest elevations as an archipelago of islands.

fig 7–9

After Oil is not a mere switch in energy producers; it is a reformation of the geographic imagination that redeems technological lands from abstraction and rescues all energy technologies from idealism and depoliticization. When energy, economy, and environment are at the forefront of design concerns today, it is important to underscore that all technopolitical projects and representations, including carbon, are simultaneously formal projects (and representations). It is a pivotal design task for energy futures to draw upon carbon's "matters of concern," looking closely at how the techno-carbon system has taken form. Moreover, a geographic imagination highlights the politics and aesthetics of all proposed technological contracts with the planet. It intervenes in representation to render visible the inequality between the promises of technological fixes and the distribution of externalities. If the transition to a low-carbon energy regime is not accompanied by reflection on the geographies on which the current regime rests and by inquiry into its proposed geographies—solar or other—future projects will simply pay lip service to an ecological discourse in urbanism rather than usher in political and economic transformation. The political success of overcoming carbon forms hinges on the reform of the geographic imagination to project a formally and politically situated vision of power in the world's sunbelts and wind corridors.

Earlier versions of this essay were published as "Where Are The Missing Spaces? The Geography of Some Uncommon Interests" in *Perspecta* 45 (2012): 109–116 and "Carbon Re-form" in *Log* 47 (2019): 106–117.

After Oil Project Credits: Rania Ghosn and El Hadi Jazairy, with Jia Weng, Rawan Al-Saffar, Kartiki Sharma, Hsin-Han Lee, Namjoo Kim, and Sihao Xiong.



fig 6 Strait of Hormuz Grand Chessboard.
Design Earth, 2016



fig 7 Bubiyan Island, There Once Was an Island. Design Earth, 2016



fig 8 Bubiyan Island, There Once Was an Island. Design Earth, 2016



fig 9 Bubiyan Island, There Once Was an Island. Design Earth, 2016

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